

# Coordinate clipping: Real Stability

Research Architecture Runtime

operator/coordinate-clipping

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## Abstract

`coordinate-clipping` uses interval clamp, which is 1-Lipschitz; hence  $|x' - x| \leq \varepsilon$  implies the clamped values move by at most  $\varepsilon$ . Lean `LEAN_FULL`.

**This theorem is:** Reduction

## 1 Problem

`coordinate-clipping` projects coordinates onto intervals via  $\text{clamp}(x; \text{lo}, \text{hi}) = \max(\text{lo}, \min(x, \text{hi}))$ .

## 2 Stability notion

Nonexpansiveness of clamp: output perturbation  $\leq$  input perturbation in absolute value.

## 3 Definitions

Require  $\text{lo} \leq \text{hi}$ . Stability form is the Lipschitz claim with  $|x' - x| \leq \varepsilon$ .

## 4 Theorem

**Theorem 1.** *If  $\text{lo} \leq \text{hi}$  and  $|x' - x| \leq \varepsilon$ , then  $|\text{clamp}(x') - \text{clamp}(x)| \leq \varepsilon$ . The Lipschitz constant 1 is sharp.*

## 5 Intuition

Clipping cannot expand distances: points move weakly toward the interval.

## 6 Examples

**Example 2.**  $\text{lo} = 0$ ,  $\text{hi} = 1$ ,  $x = 0.4$ ,  $x' = 0.7$ ,  $\varepsilon = 0.3$ : both clamp to themselves and move by 0.3.

## 7 Proof

The map  $x \mapsto \max(\text{lo}, \min(x, \text{hi}))$  is 1-Lipschitz on  $\mathbb{R}$  when  $\text{lo} \leq \text{hi}$  (standard case analysis on the three regions  $(-\infty, \text{lo}]$ ,  $[\text{lo}, \text{hi}]$ ,  $[\text{hi}, \infty)$ ). Therefore  $|\text{clamp}(x') - \text{clamp}(x)| \leq |x' - x| \leq \varepsilon$  (Theorem ??). Sharpness: take an interior segment of length  $\varepsilon$  inside  $[\text{lo}, \text{hi}]$ .

## 8 Formal statement

**Lean module**

`Research.Operators.CoordinateClipping.Preservation`

**Source file**

`lean/Research/Operators/CoordinateClipping/Preservation.lean`

**Theorems**

- `coordinate\_clipping\_clamp\_stability`
- `coordinate\_clipping\_clamp\_sharpness`

**Note**

Uses `Projection.Clamp`.

**Certificate**

`lean/certificates/coordinate-clipping/coordinate-clipping-clamp-stability`

**Status**

`LEAN_FULL` on Mathlib  $\mathbb{R}$  (`REAL_MATHLIB`)

## 9 Proof dependencies

- `Projection.Clamp` nonexpansiveness.
- Case analysis on regions.

## 10 Consequences

Interval/box/clipping operators reduce to clamp. Related: projection-interval, coordinate-clipping.